LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Tissues

Chapter / Topic: Tissues (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled diagrams wherever required.

SECTION A

6 marks

Choose the correct option.

Q1. Which tissue increases the length of the stem and the root? **[1]**

(a) Apical meristem (b) Lateral meristem
(c) Intercalary meristem (d) Parenchyma

Q2. Lignin is found in the cell walls of: **[1]**

(a) parenchyma (b) collenchyma
(c) epidermis (d) sclerenchyma

Q3. Blood is a type of: **[1]**

(a) epithelial tissue (b) connective tissue
(c) muscular tissue (d) nervous tissue

Q4. Which tissue connects one bone to another? **[1]**

(a) Tendon (b) Cartilage
(c) Adipose tissue (d) Ligament

Q5. Food is transported in plants by the: **[1]**

(a) xylem (b) epidermis
(c) phloem (d) cork

Q6. **Assertion (A):** Cardiac muscles contract and relax rhythmically throughout life.
Reason (R): Cardiac muscles are voluntary muscles. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What are the functions of stomata? Name the cells that surround them. **[2]**

OR

Why is the epidermis important for a plant? How is it adapted in desert plants?

Q8. Name the tissue that (a) stores fat below the skin, (b) smoothens bone surfaces at joints. **[2]**

SECTION C

6 marks

Q9. Name the simple permanent plant tissue that: **[3]**

(i) is made of dead cells and gives the plant strength. **[1]**

(ii) gives flexibility to parts such as leaf stalks. **[1]**

(iii) stores food and has loosely packed, thin-walled living cells. **[1]**

Q10. Describe the structure of a neuron and its function. **[3]**

OR

Give three differences between xylem and phloem.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Describe the three types of muscle tissue, with one location of each. **[3]**

(ii) Why is blood called a connective tissue? **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. As plants grow older, the outer protective tissue of the stem changes. A strip of secondary meristem replaces the epidermis, and cells cut off on the outside form a several-layer-thick cork, the bark of the tree. Cork cells are dead, compactly arranged and without intercellular spaces. Their walls contain a chemical called suberin, which makes them impervious to gases and water. **[4]**

(i) What replaces the epidermis in older stems? **[1]**

(ii) Name the chemical present in the walls of cork cells. **[1]**

(iii) How does cork protect the plant? Why is cork used to make bottle stoppers? **[2]**

~ All the best! ~